# Word Smart Flashcards (G8) — Teacher's Operational Manual

A complete guide to student webapp deployment, mobile home screen shortcuts, and real-time Google Sheets progress tracking

**Web Application Live URL:** `https://titoliviomilazzo.github.io/g8-wordsmart/`
**Target Audience:** Grade 8 (G8) ESL/EFL students & English Vocabulary Teachers
**Curriculum Base:** Word Smart (6th Edition) — 1,104 Core Vocabulary Terms (Chapters 2–12 complete)

## 1. Executive Overview & System Architecture

The Word Smart G8 Flashcard system is an independent web-based spaced-repetition training platform engineered to transition high-frequency academic vocabulary into students' long-term memory. It operates on an automated dual-layer pipeline:

- **Client Layer (Student Device):** Runs client-side in any modern mobile or desktop browser without app store installations or account sign-ups. Uses the modified SM-2 algorithm to schedule review intervals and provides two-way card flipping with immediate visual tactile feedback.
- **Sync Layer (Teacher Monitoring):** Every completed study session or session exit automatically transmits consolidated learning metrics in the background to the teacher's central Google Spreadsheet via Google Apps Script (no student login required).

## 2. Student Mobile Home Screen Shortcut Setup (PWA/Web App)

Instruct students to pin the webapp directly to their phone's home screen. It will open like a native fullscreen app without URL bars, providing seamless daily access.

### Apple iOS (Safari)

1. Open **Safari** and navigate to:
   `https://titoliviomilazzo.github.io/g8-wordsmart/`
2. Tap the **Share** button (square icon with an arrow pointing up at the bottom).
3. Scroll down and tap **Add to Home Screen**.
4. Confirm the title `Word Smart` and tap **Add**.

### Android (Chrome)

1. Open **Chrome** and navigate to:
   `https://titoliviomilazzo.github.io/g8-wordsmart/`
2. Tap the **More** menu (three vertical dots in the upper-right corner).
3. Tap **Install app** or **Add to Home screen**.
4. Tap **Add** to place the standalone shortcut on the home screen.

## 3. Student Learning Workflow

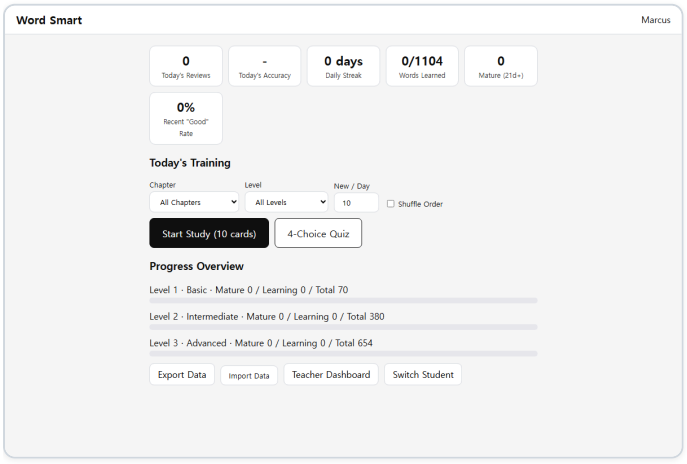



Figure 1. Student Training Portal & Configuration

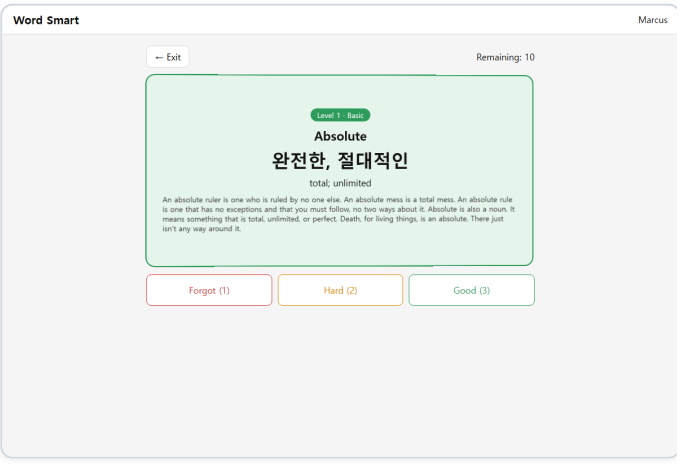



Figure 2. 3-Color Difficulty Back Face & Feedback

1. **Student Profile:** Students enter their English nickname (e.g., `David`, `Latte`, `Moka`) and select it. All personal learning curves, streaks, and card states persist locally on that device.
2. **Training Scope:** Filter by chapter (Ch 2 Core Words, Ch 4 SAT, Ch 5 GRE, Ch 9 Arts, Ch 10 Finance, Ch 11 Foreign, Ch 12 Science) and toggle **Shuffle Order** for randomized retrieval practice.
3. **Card Flip & Evaluation:**
   - Tapping anywhere on the card or pressing `Space` flips between English term and Korean definition with a smooth 3D animation.
   - Pronunciation audio can be triggered via the **Audio** button.
   - Self-rate recall: **Forgot (1)**, **Hard (2)**, or **Good (3)**. Rating triggers localized particle animations and schedules the next repetition.
4. **4-Choice Quiz:** Active recall drill where distractors are dynamically sampled from identical parts of speech to ensure rigorous discrimination.

## 4. Teacher Monitoring via Google Spreadsheet

Teachers do not need to log into the student app to verify homework or retention. The connected Google Spreadsheet automatically organizes student progress into dedicated sheets and aggregate views.

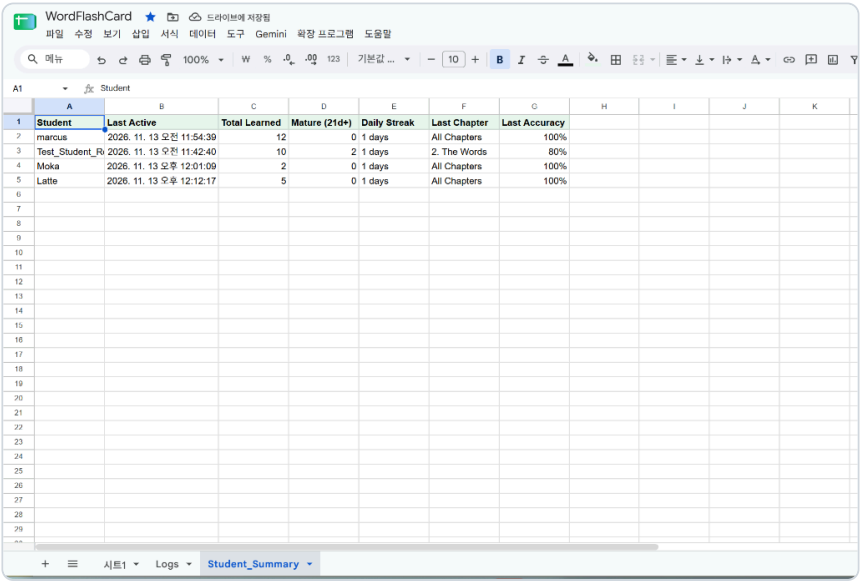



Figure 3. Central Google Spreadsheet — Automated Student Summary and Dedicated Tabs

### Spreadsheet Structure & Sheet Roles

| Sheet Name | Purpose & Data Recorded | Teacher Action |
|---|---|---|
| **Student_Summary** | Aggregated leaderboard: Last Active Timestamp, Total Learned, Mature Cards (>21d), Daily Streak, Last Accuracy. | Check overall classroom retention and verify daily homework completion at a glance. |
| **Logs** | Immutable raw session logs recording every study event timestamp, chapter, card counts, and accuracy percentage. | Audit timestamp authenticity and review study distribution over time. |
| **Dedicated Tabs** (e.g., `Latte`, `Moka`) | Auto-generated on student's first submitted session. Contains complete historical progression across dates and chapters. | Conduct individualized parent-teacher consultations and diagnose error frequencies. |

## 5. Sharing the Spreadsheet with Other Teachers

1. Open the **WordFlashCard** Google Spreadsheet.
2. Click the green **Share** button in the top-right corner.
3. Under **General access**, change setting from *Restricted* to **Anyone with the link**.
4. Set permission level to **Viewer** (recommended to prevent accidental edits) or **Editor**.
5. Click **Copy link** and share with co-teachers.

## 6. Student Management & PIN Security

- **4-Digit Student PIN:** Selecting any student profile prompts for their 4-digit PIN to prevent unauthorized access. Teachers can view and reset all student PINs in the Teacher Dashboard (Passcode: 9999) or via Google Sheets.
- **Edit Name (Edit):** Updates student nickname while preserving all card interval states, streaks, and progress.
- **Delete Student (Delete):** Prompts confirmation before purging local record.

> **Quality Specifications:** SuperMemo SM-2 Interval Calculation (21-day maturity) · UTF-8 bilingual definitions across all 1,104 entries · Non-blocking Keep-Alive background sync.